

Subject:
Physics

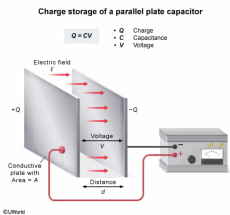
System:
Electrostatics and Circuits

An 8-F parallel plate capacitor generates a voltage of 5 V when fully charged. How much charge is stored by the capacitor?

- ☐ A. 0.62 C [4%]
- ☐ B. 1.6 C [16%]
- ☒ C. 40 C [73%]
- ☐ D. 100 C [6%]

Correct
73% answered correctly

Explanation:



A **capacitor** is a common component of electrical circuits that stores electric charge between equally but oppositely charged conductive surfaces (plates) separated by a fixed distance (d).

A capacitor is charged when it is connected to a source of electric current. As electric current flows, an electric field develops across the capacitor, which causes positive charge to accumulate on the plate of the capacitor closest to the positive (+) node of the current source. Conversely, negative charge accumulates on the capacitor plate closest to the negative (–) node.

Storing equal but opposite charges in proximity to one another generates an **electric potential energy** (voltage) from the mutual attraction between the positive and negative charges. Because movement of charge between the plates (leakage) would diminish the stored charge, capacitor strength is assessed by the quantity of **charge (Q)** that can be stored relative to the **voltage (V)** generated. As such, capacitor strength is quantified by **capacitance (C)**, which is expressed through the equation:

$$C = \frac{Q}{V}$$

Capacitance is generally stated in units of farads (F), where 1 farad is equal to 1 coulomb (C) of charge stored per volt (V) of electric potential generated.

In this question, an 8-F parallel plate capacitor generates 5 V of electric potential by storing an unknown quantity of charge. To identify the amount of charge stored, the equation for capacitance must be rearranged as:

$$Q = C \times V$$

Substituting the values for capacitance and voltage into the equation gives:

$$Q = (8 \text{ F}) (5 \text{ V}) = 40 \text{ C}$$

Therefore, an **8-F parallel plate capacitor** that generates 5 V of electric potential has stored **40 C of charge (Choice C)**.

Educational objective:

Capacitors generate electric potential (voltage) by storing charge. The quantity of charge stored by a capacitor is a product of capacitance (capacitor strength) and the voltage generated by charge storage.